

The data doesn't match the task. So we built the missing half.

THE GAP THAT DEFINED THE BUILD

What we were given vs what we are scored on

	TRAINING DATA	SCORED ON
Videos	3,173 short clips, 5–30 s	240–629 s
Events	exactly one per clip	up to 6, mixed classes
Timestamps	event $\equiv$ whole clip for 7 of 11 classes	gated at tIoU $\geq$ 0.5
Marks	D1 = 25	D2 + D3 = 75

75 of the 100 marks had no training data at all.

WE MEASURED THE SHORTCUT WE CREATED

Concatenating clips teaches the model to detect scene cuts

A model trained on spliced video learns **event onset $\equiv$ cut**: brilliant on synthetic data, collapsing on real footage, which has no cuts. We censused pre-event footage per class first — **6 of 11 classes have literally none**.

0.536 is near the honest floor given those 6 classes. **We report it rather than hide it.**

THE DEMO · [DEMO/INDEX.HTML](#)

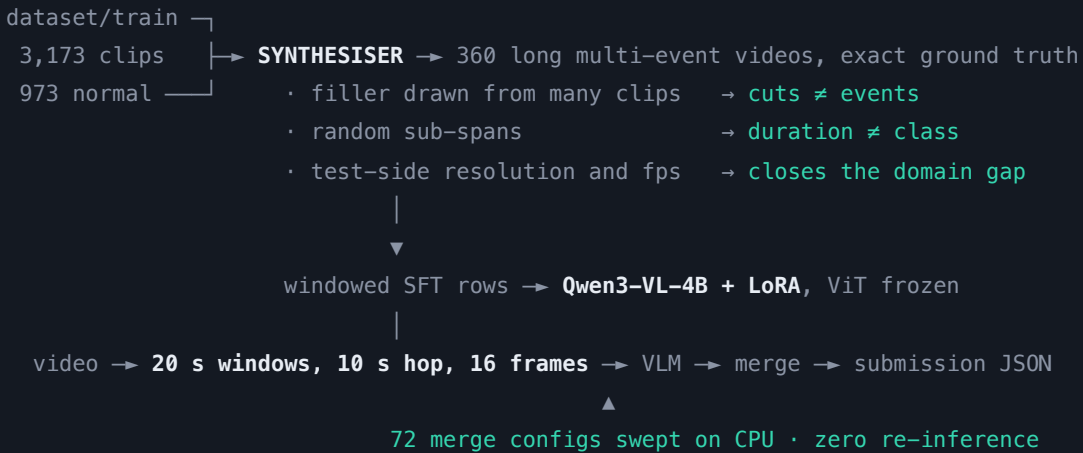
It replays each prediction on the video's clock and scores itself live



Truth above, model below, joined by a ribbon — a **perfect match is a rectangle**, a **timing error a visible skew**. It imports the same matcher that scores the run, so the page cannot disagree with the score.

PIPELINE

Everything cacheable is cached, so tuning costs no GPU



MEASURED COST – THE REAL-TIME CLAIM

One A100-40GB, 34 videos, 56 minutes of footage

326	5,216	1,830	5.5x	2.8x
WINDOWS	FRAMES	MS / WINDOW	VS REAL TIME	AT P95

A new window closes every **10 s** of video and is scored in **1.83 s**. The system keeps up with **5.5× headroom**, and still 2.8× at the 95th percentile — so real-time holds on the tail, not just on average.

MODEL SELECTION

Five VLMs, identical 24-video D1 set, same prompt and policy

MODEL	P	R	F1	SILENT
Qwen3-VL-4B CHOSEN	0.77	0.50	0.61	8/24
Qwen3-VL-8B	0.62	0.40	0.48	8/24
Cosmos-Reason2-8B	0.57	0.20	0.30	13/24
Qwen3-VL-2B	0.30	0.15	0.20	12/24
Qwen2.5-VL-7B	0.00	0.00	0.00	20/24

The 4B and 8B are **statistically indistinguishable** (slide 2). We ship the 4B: **half the size at equal accuracy.**

Every failure that mattered was silent.

So we validated artifacts, not outputs.

All numbers measured on the 34-video public set
24 videos, 20 positives → ±1 TP ≈ 5 pp of recall

1 · THE PATTERN

Seven silent failures in seven hours. None crashed.

#	FAILURE	WOULD HAVE READ AS
1	ffmpeg -t silently returns a shorter segment	later timestamps wrong
2	49 of 2,092 train rows have end ≤ start	garbage spans
3	Clip pool keyed on folder , not class label	normal traffic taught as wrong-way
4	Empty output ≡ correct "no events"	a parser bug scoring clean
5	One bad API response cut a 34-video run to 6 — exit code 0	a complete run, real score
6	Buffered output hid a 20-minute GPU job	a hang
7	The fix for #5 caused #7: error handling wrote 10 failures as "events": []	a model that is just bad

All seven would have read as "the model just isn't very good." #7 was caught only by the diagnostic added for #4 — recording raw model text on every empty result. Each is now a test.

2 · WE RETRACTED OUR OWN HEADLINE

Before it shipped, because someone asked for the mechanism

~~Bigger is worse: Qwen3-VL 4B beats the 8B, F1 0.61 vs 0.48.~~

VIDEO	TRUTH	4B	8B
T007	traffic_accident	correct	silent
T011	vehicle_blocking	silent	smoke
T014	smoke	correct	fire

The two models differ on **three videos out of twenty-four**. Sign test on 2 informative trials: **p = 0.25**. The whole F1 gap is 10 TP vs 8 TP — inside the ±1 TP noise band we had been telling everyone to apply, and then failed to apply ourselves.

Checked before retracting: both emit clean JSON, zero truncation, and the **same** number of empty windows — equally willing to answer.

Corrected claim: the 4B and 8B are indistinguishable here. We ship the 4B because it is half the size at equal accuracy.

5 · A NEGATIVE RESULT, TESTED PROPERLY

The ego-motion gate every paper implies

Every paper in the supplied reading list evaluates on **fixed-camera CCTV**; a third of this data is drone and a third dashcam. Frame differencing assumes a still background. H.264 **motion vectors are free out of the decoder** — fit a global affine with RANSAC, and the residual is independently-moving objects.

0.556	0.500
AUC, IN-EVENT VS NOT	CHANCE

Against a 2,000-trial permutation null, only **one of six tested thresholds** separates from chance (1.35× lift, p≈0.06 corrected). **It does not work**. So the cheap always-on gate in a cascade cannot be built this way — a real architectural constraint, settled in **1,165 CPU-seconds and no GPU**.

What survived: ego-motion itself, free and per-second, recovering the domain field the manifest ships **empty for all 34 videos** — 15 videos barely move, 6 move part of the time, 13 move throughout.

3 · CHEAP POLICY BEAT MODEL CHANGES

Both free, both CPU-side, no re-inference

D1 OUTPUT POLICY	TP	FP	FN
every predicted event	10	12	10
top-1 per video	10	3	10

False alarms cut 4× at zero recall cost. A D1 video has at most one truth event, so the extras could only ever have been false alarms. One entrant on this leaderboard lost **26 of 35 marks to 94 false alarms**. Does **not** transfer to D2/D3, where it deletes real detections.

And the obvious fix that does **not work** — confidence carries no signal

D2/D3 CONFIDENCE GATE	TP	FP
≥ 0.5	1	31
≥ 0.7	1	31
≥ 0.9	1	29

0.5 → 0.9 removes **two** false alarms out of 31. Whatever gates them must come from **outside** the model's own score.

6 · WHERE WE ACTUALLY ARE

First scored upload, and the arithmetic that reframes it

	SCORED	OF	MARKS / VIDEO
D1 · 24 short clips	12.1	25	1.04
D2 · 6 × 240 s	11.7	35	5.83
D3 · 4 long videos	0.0	40	10.00
Total	23.8	100	

D2's 11.7 is **exactly the empty-submission floor** — we earned nothing there yet. And **one D3 video is worth ten D1 videos**, while we sit at zero on D3. The whole remaining opportunity is long-video temporal localisation, which is precisely the half the training data never contained.

Controlled runs killed two of three explanations for the model's silence: more frames does not help (24 and 32 are worse than 16), and neither prompt variant helps. **Five interventions, zero improvements** — leaving fine-tuning as the only lever, which is where slide 1's synthesised data goes.